

Program : Diploma in Biomedical Engineering	
Course Code : 5243B	Course Title: Programming in C
Semester : 5	Credits: 4
Course Category : Program Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester : 60

Course Objectives:

- To provide a basic knowledge in C Programming.
- To enhance programming skills using Control blocks, Looping, String Manipulation, and functions to solve real world problems.
- To make capable to solve real-life programming tasks using C.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic principles and theorms of Engineering Mathematics		Mathematics I &II	1 & 2
Basic functions and features of Computer, Operating system and Internet applications, basic programming skills in Python.		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Make use of programming concepts with C programming.	13	Applying
CO2	Make use of iterative control structures and arrays in programs.	15	Applying
CO3	Develop programs using pointers and strings to solve problems more efficiently.	15	Applying
CO4	Make use of functions to solve problems effectively	15	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Make use of programming concepts with C programming.		
M1.01	Understand structure of a C program	1	Understanding.
M1.02	Use the terms Keywords, Variables, Constants, Data types and type qualifiers	3	Applying
M1.03	Identify output and input functions	3	Applying
M1.04	Identify operators	4	Applying
M1.05	Identify type casting	2	Applying
Contents: Introduction to Programming concepts: Structure of a C program - Keyword, Variables, Constants, Data types and type qualifiers. Output and input functions, Operators - Arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, Assignment, Conditional and type casting.			
CO2	Make use of iterative control structures and arrays in programs.		
M2.01	Identify the control structures – if, if-else nested if and switch	2	Applying
M2.02	Identify loop structures – while, do-while, for and nested loops.	2	Applying
M2.03	Make use of unconditional control statements – break, continue and goto	2	Applying
M2.04	Identify one dimensional array operations – insertion, deletion, searching, sorting	3	Applying.
M2.05	Use two dimensional array deceleration and matrix operations	3	Applying.
M2.06	Make use of matrix manipulations	3	Applying
	Series Test – I	1	

Contents:

Iterative control structures and arrays: Control structures - if, if-else, nested if statement, and switch. Looping structures - while, do-while, for, nested loops, Unconditional control statements-break, continue, goto.

Array operations - insertion, deletion, searching, sorting. Two dimensional array declaration - two dimensional array operations - Matrix manipulations (addition, multiplication etc)

CO3	Develop programs using pointers and strings to solve problems more efficiently.		
M3.01	Use pointer concepts in programming	2	Applying
M3.02	Identify pointer arithmetic operations - addition, subtraction, increment, decrement and comparison.	3	Applying
M3.03	Develop programs for accessing strings using pointers	3	Applying
M3.04	Utilize gets() and puts() functions	3	Applying
M3.05	Use built in string handling functions like strcpy(), strlen(), strcat(), strcmp(), strcmp()	4	Applying

Contents:

Pointers and string operations: Basics of Pointers, Pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and Puts() functions, String handling functions.

CO4	Make use of functions to solve problems effectively		
M4.01	Explain functions, types and advantages	2	Understanding
M4.02	Illustrate structure of a user defined functions	3	Understanding
M4.03	Make use of function prototype, function call, arguments, return type	3	Applying
M4.04	Identify different types of functions	2	Applying
M4.05	Make use of call by value and call by reference	3	Applying
M4.06	Make use of recursive functions	2	Applying
	Series Test – II	1	

Contents:

Functions: Definition of Functions - Structure of user defined functions - functionprototype, function call, arguments, return type - Passing arguments to a Function: call byvalue, call by reference, Recursive functions.

Text / Reference:

T/R	Book Title/Author
T1	Programming in ANSI C by Balaguruswami E, TMH publications
R1	Computing Fundamentals and C Programming 2nd Edition by E Balaguruswami, TMH publishers
R2	Programming in C - by ReemaThareja, OUP India publishers
R3	C Programming Absolute Beginner's Guide by Greg Perry, Dean Miller, QUE Publications
R4	C: The Complete Reference by Herbert Schildt, TMH Publications
R5	Programming in C - second edition - R. Subburaj, - Vikas Publishing House.

Online Resources:

Sl.No	Website Link
1	https://www.programiz.com/c-programming
2	https://www.tutorialspoint.com/cprogramming/index.htm
3	https://www.cprogramming.com/
4	https://www.geeksforgeeks.org/c-programming-language/
5	https://www.freecodecamp.org/news/the-c-beginners-handbook/